PROGRAMMING IN JAVA

CO1 AT-3 : PSEUDOCODE WRITING

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Q1 **Student Management System [BL3 – Apply]** 10 marks

Write pseudocode to design a Student class with attributes (name, rollNo, marks) and methods to:
• Accept student details
• Calculate average marks
• Display result (Pass/Fail)
Apply encapsulation and data hiding by restricting direct access to attributes.
OOP Concepts: Encapsulation | Data Hiding | Classes & Objects

Your Answer

```
START
CLASS student
PRIVATE:
    name,rollno,marks
END MENTHOD
METHOD acceptdetails()
```

Save Answer

Q2 **2. Library Book Management [BL3 – Apply]** 10 marks

Design a Book class with private attributes (title, author, ISBN, isAvailable). Implement methods to:
• Issue a book (change availability status)
• Return a book
• Display book details using getter methods
Demonstrate object creation for at least 3 books and simulate issue/return operations.
OOP Concepts: Encapsulation | Getters & Setters | Objects

Your Answer

```
START
CLASS book
PRIVATE:
    title,author,ISBN,isavailable
PUBLIC:
METHOD issuebook()
```

Save Answer

Q3 **Employee Salary Calculation [BL3 – Apply]** 10 marks

Write pseudocode for an Employee class with encapsulated attributes (empId, name, basicPay, designation). Implement methods to: • Calculate gross salary (basic + HRA + DA)
• Calculate net salary after deductions (PF, tax)
• Display pay slip
Apply data hiding so salary components are not directly accessible.
OOP Concepts: Encapsulation | Data Hiding | Methods

Your Answer

```
CLASS employee
PRIVATE empid,name,basicpay,designation
PRIVATE hra,da,pf,tax

METHOD calculategross()
  gross=basicpay+hra+da
```

Save Answer

Q4 **Shape Area Calculator using Polymorphism [BL3 – Apply]** 10 marks

Create a Shape base class with an overridable calculateArea() method. Apply this to:
• Circle — area using radius
• Rectangle — area using length and breadth
• Triangle — area using base and height
Use method overriding to demonstrate runtime polymorphism. Call calculateArea() for each object.
OOP Concepts: Inheritance | Polymorphism | Method Overriding

Your Answer

```
START
CLASS shape
   METHOD calculatearea()
      area=3.14*r*r
END CLASS
CLASS rectangle EXTENDS shape
```

Save Answer

Q5 **Vehicle Registration System [BL3 – Apply]** 10 marks

Design a Vehicle class with attributes (regNo, owner, fuelType). Extend it into:
• TwoWheeler — with engine capacity
• FourWheeler — with seating capacity and AC status
Apply inheritance to reuse common attributes and override a displayDetails() method in each subclass.
OOP Concepts: Inheritance | Method Overriding | Subclass

Your Answer

```
START
CLASS vechicle
   regno,owner,fueltype
METHOD displaydetails()
END CLASS
CLASS twowheeler EXTENDS vechicle
```

Save Answer

Q6 **Bank Account Hierarchy [BL4 – Analyze]** 10 marks

Develop pseudocode for a BankAccount superclass and two subclasses — SavingsAccount (with interest calculation) and CurrentAccount (with overdraft facility). Analyze and implement:
• Inheritance of common attributes (accNo, balance, holder)
• Method overriding for withdrawal behavior
• How overdraft protection differs from savings limit
Compare the behavior of withdraw() across both subclasses and justify the design choices.
OOP Concepts: Inheritance | Method Overriding | Polymorphism

Your Answer

```
START
CLASS bankaccount
    accno, balance, holder
    METHOD withdraw()
END CLASS
CLASS savingsaccount EXTENDS
```

Save Answer

Q7 **Hospital Patient Record System [BL4 – Analyze]** 10 marks

Design a Patient base class and analyze how it extends into:
• InPatient — with ward number, admission date, and daily charges
• OutPatient — with appointment date and consultation fee
Analyze encapsulation requirements for sensitive data (diagnosis, prescription). Override a generateBill() method for each type and compare the billing logic differences. OOP Concepts: Encapsulation | Inheritance | Polymorphism

Your Answer

```
i.generatebill()

CREATE outpatient o
o.generatebill()

END
```

Save Answer

Q8 **E-Commerce Product Catalog [BL4 – Analyze]** 10 marks

Analyze and design a Product class hierarchy for an e-commerce system:
• Electronics — with warranty, brand, voltage
• Clothing — with size, fabric, gender
• Grocery — with expiryDate and weight
Override an applyDiscount() method for each category with different discount rules. Analyze why polymorphism makes the cart checkout process flexible and extensible. OOP Concepts: Polymorphism | Inheritance | Method Overriding

Your Answer

```
START
CLASS product
        productid,name,price
METHOD applydiscount()
END CLASS
CLASS electronics EXTENDS product
```

Save Answer

Q9 **University Staff Hierarchy [BL4 – Analyze]** 10 marks

Analyze the design of a Staff base class extended by TeachingStaff and NonTeachingStaff. Further extend TeachingStaff into Professor and Lecturer (multilevel inheritance). For each level:

• Identify which attributes should be private vs protected

• Override calculateAllowance() at each level

• Analyze how protected access enables inheritance without full exposure Justify the use of multilevel inheritance and explain how access modifiers enforce data hiding.

OOP Concepts: Multilevel Inheritance | Access Modifiers | Data Hiding

Your Answer

```
p.calculateallowance()
CREATE LECTURER 1
1.calculateallowance()
CREATE nonteachingstaff n
n.calculateallowance()
END
```

Save Answer

Q10 **Smart Home Device Controller [BL4 – Analyze]** 10 marks

Analyze and design a SmartDevice superclass with subclasses: SmartLight, SmartThermostat, and SmartLock. Override an executeCommand(cmd) method in each, where the same command (e.g., 'on', 'off', 'status') produces device-specific behavior. Analyze:

• How polymorphism allows a single controller loop to manage all devices

• Security implications of data hiding in SmartLock (PIN must never be exposed) Design the class hierarchy and explain the OOP principles applied at each level.

OOP Concepts: Polymorphism | Encapsulation | Inheritance

Your Answer

```
START
CLASS smartdevice
    devicename
METHOD executecommand(cmd)
END CLASS
CLASS smartlight EXTENCDS smartdevice
```

Save Answer